Fundamental Labs 1 Chapter 11 DC motor manufacturing and control

Report

Introduction

1.1 Purpose of this lab

To build and rotate a motor, and thereby

Learn about system assembly and the use of tools and measuring instruments.

Understand a DC motor's operating characteristics and feedback control system behavior.

Understand the need for and usefulness of electromagnetism, circuit logic, control theory, etc.

1.2 Lab summary

First week

Explanation, build a brush-type DC bipolar motor.

Second week

Brushless motor, velocity characteristics measurement. Configuration of feedback-based control system.

Attempt to use a motor as a generator (observe operation).

A. First week (1/2): Bipolar motor building, Brush DC motor

Assignment K1 Brushed direct current motor

1. Draw and explain four figures showing why such direct current motors rotate, and explain the role of the brushes.
   There are 4 stages in rotating the DC motor. The four figures are the following below, with each figure representing a different stage
   State 1:

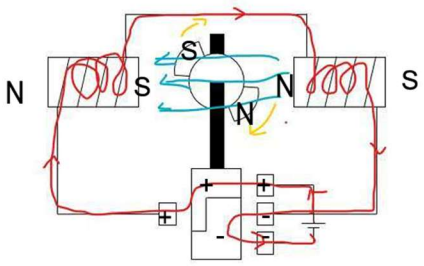
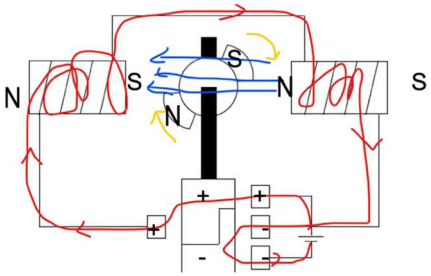


Figure 1

The current flows from the battery through the brushes, the rotor, the two electromagnets and back to the negative terminal of the battery. It creates a magnetic field between the two electromagnets as shown in the picture above. The permanent rotor is situated with the opposite polarity, so it repels the electromagnets causing the rotors to rotate.

State 2:



In this state the current flows in the same direction as in state 1. However, the rotor is in the opposite direction, and the force on the rotor is attractive and the rotor continues to rotate until its field aligns with the electromagnet.

Figure 2

State 3:

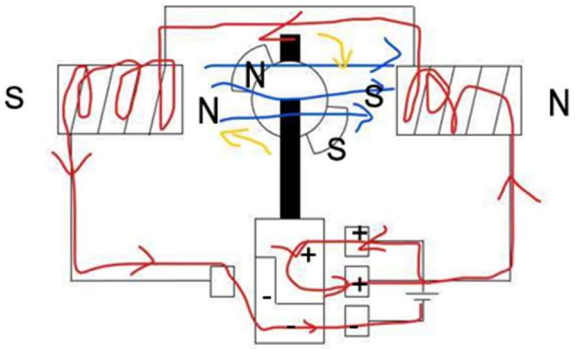


Figure 3

In this state, the current flows in the opposite direction from stage 3. This creates a magnetic field in the opposite direction. The magnetic field of the permanent magnet in

the rotor is in the opposite direction of the electromagnets, so it repels the field and rotates in the direction shown by the yellow arrow.

State 4:

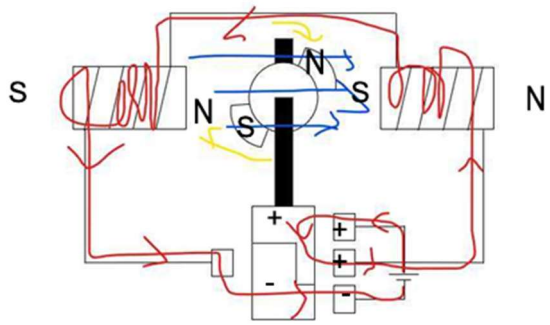


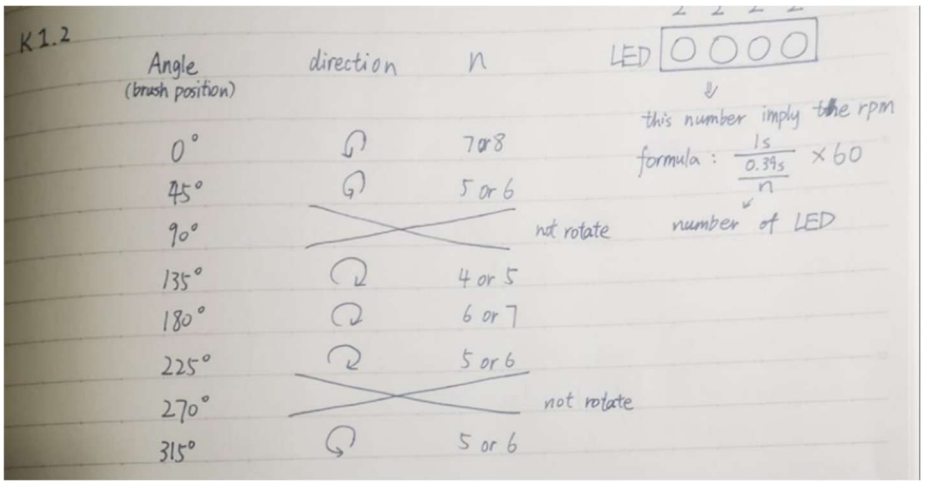
Figure 4

The current flow in this state is the same as in State 3, and thus the magnetic field is the same. But the rotor is in the opposite direction to State 3. The magnetic field of the rotor exerts an attractive force on the electromagnets this time, causing it to continue rotating in the direction shown.

From these stages we can see how the brushed rotor operates. The role of the brushes is to switch the direction of current flow to alternate the magnetic field every time the rotor rotates 180 degrees.

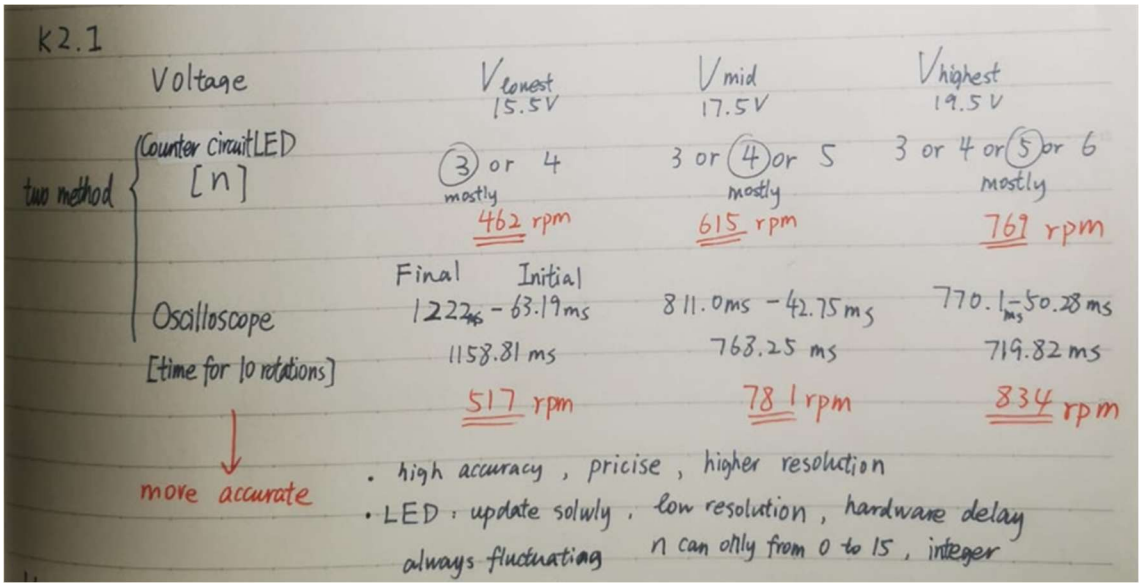
2. When brush timing adjustment (specifically, adjusting by loosening hexagon socket set screws and changing the position of rotor notches) is done, for what positional relationship of the rotor and magnets does the motor rotate better? Record.

Our team has recorded the following data



K2 Enter the rotational speed measurement results for the brushed motor in Section 3.3 using each method, and then explain the following items in the results.

1. Of the rotational speeds determined from the oscilloscope observation results and the rotational speed acquired from the motor control board LED display, which rotational speed appears to be more accurate? Indicate the reason?



Based on our team's results, the rotational speeds obtained from the oscilloscope method is more accurate than that acquired from the LED display. These is because the oscilloscope has high accuracy and higher resolution. Whereas the LED updates slowly, has low resolution, may has hardware delay, and always fluctuating.

2. How may the accuracy of the rotational speed measurements using the counter circuit on the motor control board be improved?

   The accuracy may be improved by increasing the clock frequency, decreasing the sampling interval and when calculating, use the weighted average.

B. Second week (1/3): Brushless motor, velocity characteristics measurement

Assignment K3 First, adjust the power supply voltage to 5 V. Manual External input 1. Observe with the oscilloscope how the direction of the voltage output by the coil changes according to the position of the light shielding plate while slowly rotating the motor by hand, and in Fig. 10, observe the potential between the DC motor control board terminals A and GND, and furthermore between B and GND.

Summarize in a table the relationship between the light shielding plate position and low or high potential in terminals A and B from the measurement results.

The light shielding plate	A(V)	B(V)	Direction of the current
Covering the photo interrupter	58.96 mV	4.51 mV	B → A
Not	4.501 mV	192.4 mV	A → B

2. Rotate the brushless motor, and describe in your report the role played by the light shielding plate when rotating the brushless motor.

The light shielding plate changes the direction of the motor when it rotates.

3. In addition, study why a brushless motor rotates more efficiently than a brushed motor, and explain this in your report

1) Higher Efficiency: A brushless motor rotates more efficiently than a brushed motor because a brushless motor has less friction and thus less energy loss. The brush in the motor gives friction. In a brushless motor, we do not need to worry about the friction loss against the brush. One of the more significant advantages of brushless DC motors is that they are between 85% and 90% more efficient than brushed motors which only have an 80% efficiency rate. For the most part, this high efficiency comes from the reduction of sound and heat, which in turn minimises loss of power.
2) Easier to control and has precise speed control: the motor's brushless design ensures that the unit can alternate between low speeds and high speeds.

3) Longer Lifespan: Brushless DC electric motors can easily last up to 10,000 running hours provided the application remains within the manufacturer's recommended guidelines for continuous operations. Since brushless DC motors have fewer moving parts, less components can wear out over time.

4) High Torque-to-Weight Ratio: With fewer major components, brushless motors offer a stronger torque-to-weight and power-to-weight ratio than brushed motors. Less energy is wasted in friction between the commutator and brushes (as in the case of a brushed motor). When this happens it generates torque and much higher efficiency.

5) Wide Operating Voltage Range : Being able to operate over a wider range of voltage levels means that these motors can be used with greater flexibility in applications that require varying power options.

6) Low Maintenance : Unlike brushed DC motors, brushless motors don't have carbon brushes that need to be continuously replaced. This drastically reduces the maintenance requirements. Not only does this mean that brushless DC motors face less downtime, but the maintenance costs are minimal.

7) Smoother Operation : With fewer components, there's consistent and smoother overall operation.

8) Improved Heat Dissipation : A lack of brushes ensures that BLDCs distribute heat more evenly and this ultimately reduces the risk of overheating. This stems mostly from less friction between components.

Reference:

46 Brushless DC Motor Advantages Over Brushed (Ultimate Guide)

https://assunmotor.com/blog/brushless-dc-motor-advantages/

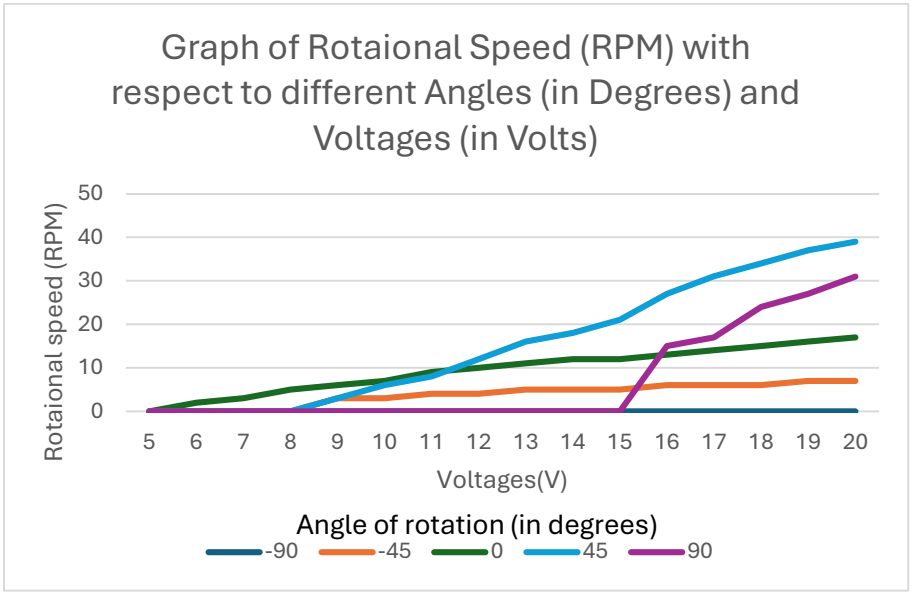
Assignment K4 Try to determine the relationship between the motor rotational speed and the motor supply voltage.

1. Assuming the position of the light shielding plate found in 4.3 to be 0°, change the position of the light shielding plate in 45° increments from -90° (assuming the motor rotating direction to be +) to +90° (change the switch timing of the direction of current flowing to the motor), determine the rotational speed from the voltage at which the motor starts to rotate (confirm using a power supply output voltage meter) in 1-V increments, and record the rotational speed (record when the motor does not rotate as 0 rotations). Then, draw a graph showing measurement results. Furthermore, the only good method for rotational speed measurement is to use the rotational speed display LED on the control board (In this case, the motor sometimes does not rotate when the voltage is low, so set a high voltage and determine the voltage at which the motor starts to rotate).

The rotational speed with respect to different angles of rotation (measured in degrees) and the voltage values (measured in volts or V), are recorded in the table below.

Angle rotation(in degr		-90	-45	0	45	90
Voltage(V)	5	0	0	0	0	0
	6	0	0	2	0	0
	7	0	0	3	0	0
	8	0	0	5	0	0
	9	0	3	6	3	0
	10	0	3	7	6	0
	11	0	4	9	8	0
	12	0	4	10	12	0
	13	0	5	11	16	0
	14	0	5	12	18	0
	15	0	5	12	21	0
	16	0	6	13	27	15
	17	0	6	14	31	17
	18	0	6	15	34	24
	19	0	7	16	37	27
	20	0	7	17	39	31

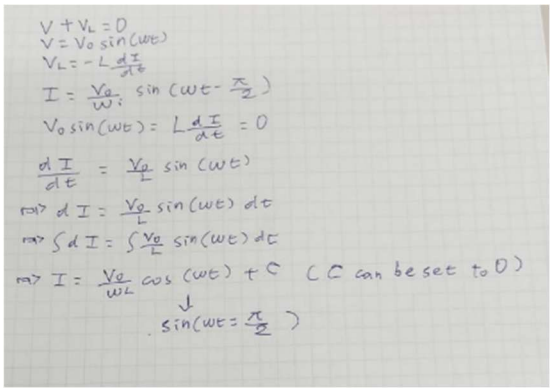
The graph showing the relationship between rotational speed (in RPM) to the angle of rotation and voltage values are below in the line chart



The different colour lines are according to each angle of rotation

2. The rotational speed will theoretically reach its maximum at +90° in the lab results. Examine the relationship between the voltage values, rotational velocity, and the light shielding plate angle, and express it in an equation (hint: What is the induced power from coil self-inductance? Does the current direction simultaneously change with the timing by which voltage is switched?). In addition, discuss whether the lab results are in agreement with theoretical values.

The relationship between the voltage values, rotational velocity, and the light shielding plate angle are expressed in this equation with L = inductance, I= current and V_0 and V_L = rotational speed



Here, the value of sin (ωt) is maximum when the angle is π/2 radian, or 90 degrees.

3. Did the optimal rotor angle of the brushed motor in K1-2 equal the angle of the light shielding plate at which the rotational speed was at its maximum in K4-1? If not, explain why.
   The optimal rotor angle of the brushed motor in K1-2 is 0°, while the angle of the light shielding plate at which the rotational speed was at its maximum in K4- is 90°. For assignment K1-4, this is because the equations, which is dependent on sine, is maximum when the angle is 90°.

C. Second week (2/3): Feedback-based control system configuration

Assignment K5 Feedback control

1. The size of the voltage provided to the power supply input terminals is proportional to the value shown by the LED to the bottom right of the control board (binary display). Use an oscilloscope to confirm that the voltage supplied from the power supply output terminals changes when the values shown by the LED change.

2. Set two types of target rotational speed (high rotational speed and low rotational speed), and record how the rotational speed and voltage changes with time every 10 s immediately after setting. The rotational speed at the start of the measurement starts from a value higher than the target when the target is low, and lower than the target when the target is high. (Note 1) Sometimes the motor suddenly stops before reaching the set rotational speed, but this is because the control board limits the voltage that can be output. In such cases, reduce the target rotational speed and run the experiment again. (Note 2) Retrieve a minimum of 2 min of data after the motor has settled at

its target rotational speed, so that it is possible to confirm a stable rotational speed state.

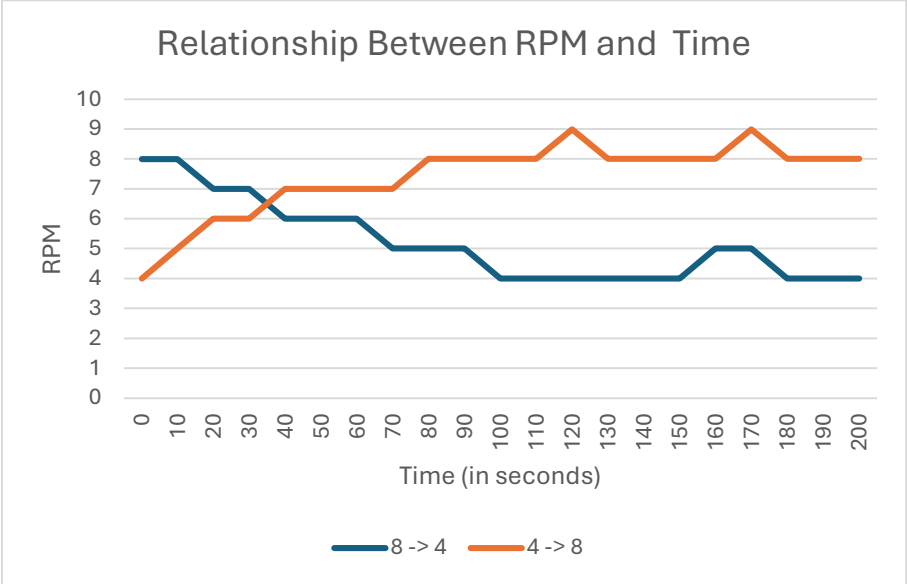
A) The voltage values recorded every 10 seconds are below, one is from high value to low target and the other is low value to low target

Time	0	10	20	30	40	50	60	70	80	90	100
8 -> 4	12.5	12	11	11	10.5	10	9.5	9	8.5	8	7.7
4 -> 8	7.5	7.6	8.5	9	9.5	10	10.5	11	11.5	11.8	12

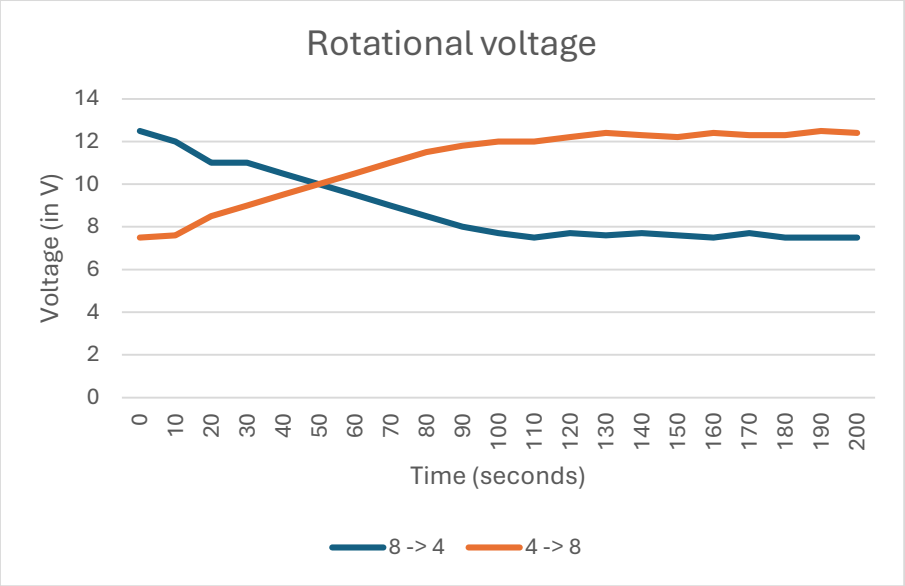
B) The rotational speed (in RPM)

142.8571	0	10	20	30	40	50	60	70	80	90	100
8 -> 4	8	8	7	7	6	6	6	5	5	5	4
4 -> 8	4	5	6	6	7	7	7	7	8	8	8

3. Create graphs from the above two measurement results for the relationship between the time and rotational speed, and between time and voltage
   a) The graph of the relationship between the time and rotational speed



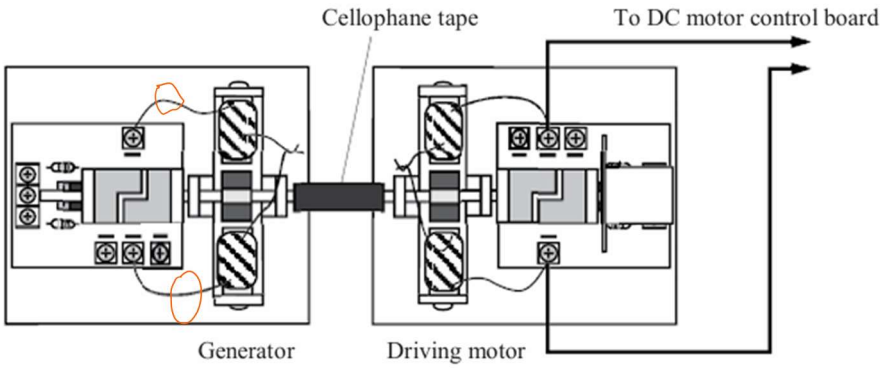
   b) The graph of the relationship between the time and Voltages



D. Second week (3/3): Using the motor as a generator

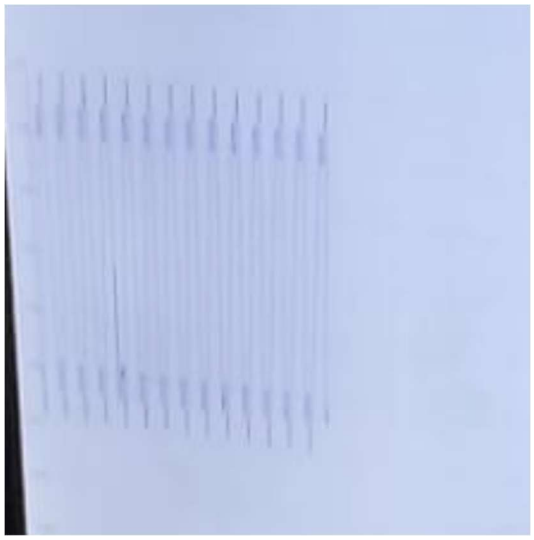
K6 How should the terminal be observed in order to observe the induced voltage? Explain in the report using figures.

The terminals should be connected to the two electromagnets to measure the voltage



K7 Save the generator oscilloscope waveforms observed for three rotational speeds, and confirm each rotational speed and induced voltage.

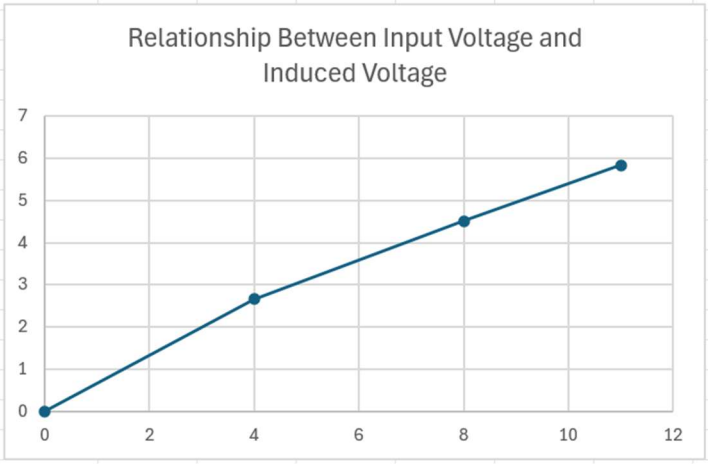
The oscilloscope showing the results are due tomorrow



The oscilloscope shows sinusoidal graph for the brushless motor, whereas for the brush motor, the graph is not sinusoidal. Thus the values for the brushless motor are larger.

K8 Try to plot the relationship between the rotational speed and induced voltage of the generator waveform observed. Explain the meaning of the graph. Discuss by referring to the previously learned law of electromagnetic induction.

InputVolta	10	15	20
Voltage	2.66	4.51	5.84
LED	4	8	11
	0	0	
	4	2.66	
	8	4.51	
	11	5.84	
	-->Linear		



From the graph we can see that the graph has a linear relationship due to the proportionality in Faraday's law. The induced voltage (EMF) is proportional to the rate of change of magnetic flux through the coil.